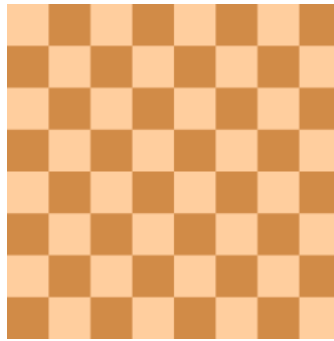


Introduction to Computing: Tutorial 4

BSDS I Year, 2026–2027

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1. Consider a chessboard as shown in the figure below.



For simplicity, assume that the squares are numbered $(0, 0), (0, 1), \dots, (0, 7), (1, 0), (1, 1), \dots, (7, 7)$, starting from the top left square, proceeding left-to-right within each row, then row-by-row from top to bottom, and ending at the bottom right square.¹

You are given the positions of two pieces of the same kind. The pieces are either both rooks (permitted to move any number of steps either horizontally or vertically), or both bishops (permitted to move any number of steps diagonally), or both queens. Write a Python function to determine whether the pieces are in mutually attacking positions. Assume that there are no other pieces on the board.

Input parameters: two 2-tuples, specifying the positions of the pieces, and a string, specifying the type of the piece ("rook", "bishop", or "queen").

Return value: `True` or `False`

[Think about how the function would need to be modified if the pieces were not of the same type, and if other pieces were present on the board at known positions).]

2. Write a function `longest_common_prefix(*args)` that takes any number of strings as arguments, and returns the longest prefix that is common to all the strings. For example,
 - `longest_common_prefix("hat", "hatch")` should return "hat",
 - `longest_common_prefix("class", "cancelled")` should return "c", and
 - `longest_common_prefix("apple", "orange", "banana")` should return the empty string.

If no strings are passed, your function should return the empty string. What should your function return if only one string is passed?

¹This is called *row-major order*.

3. Write a function `rh_derivative(f, x)` to approximately compute $f'(x)$ using the following:

$$f'(x) \approx \frac{f(x+h) - f(x)}{h} \quad (h > 0).$$

Choose a sufficiently small value of h . There are no formal requirements; any ‘reasonable’ approximation will do, i.e., the computed value should be close to the actual value in your judgment. Test your function on functions from the `math` package; remember to add `import math` at the beginning of your program.